

Caterpillar is leading the power generation marketplace with Power Solutions engineered to deliver unmatched flexibility, expandability, reliability, and cost-effectiveness.



Specifications

Generator Set Specifications		
Minimum Rating	1230 ekW	
Maximum Rating	1500 ekW	
Emissions/Fuel Strategy	Low Fuel, Low Emissions	
Voltage	380 to 13800 Volts	
Frequency	60 Hz	
Speed	1800 rpm	
Duty Cycle	Standby, Mission Critical, Prime, Continuous	

Engine Specifications		
Engine Model	3512B TA, V-12, 4-Stroke Water-Cooled Diesel	
Bore	170 mm	6.69 in
Stroke	190 mm	7.48 in
Displacement	58.56 l	3573.55 in
Compression Ratio	14.0:1 Std 15.5:1 HD	
Aspiration	TA	
Fuel System	Electronic Unit Injection	
Governor Type	Adem™3	

Generator Set Dimensions		
Length - Maximum	5637.5 mm	221.9 in
Width - Maximum	2286 mm	90 in
Height - Maximum	2367.2 mm	93.2 in
Dry Weight - Genset (minimum)	25090 kg	24430 lb
Dry Weight - Genset (maximum)	11380 kg	25090 lb

Benefits and Features

Cat Generator Set Package

- Cat generator set packages have been fully prototype tested
- Accepts 100% block load in one step and meets NFPA 110 loading requirements
- Conform to ISO 8528-5 steady state and transient response requirements

Cat Diesel Engine

- Reliable, rugged, durable design
- Field-proven in thousands of applications worldwide
- Four-stroke-cycle diesel engine combines consistent performance and excellent fuel economy with minimum weight

SR5 Alternator

- Superior motor starting capability minimizes need for oversizing alternator
- Designed to match performance and output characteristics of Cat diesel engines
- Robust Class H insulation

EMCP 4 Control Panel

- User-friendly interface and navigation
- Scalable system to meet a wide range of installation requirement
- Expansion modules and site-specific programming for specific customer requirements

Cooling System

- Designed to operate in standard ambient temperatures up to 50°C (122°F)
- Contact your Cat Dealer for specific ambient and altitude capabilities

Certifications

- UL 2200 Listed
- CSA
- IBC seismic certification
- OSHPD pre-approval

One Safe Source

- Components used in the generator set are selected at the design stage to provide optimum package performance
- The generator set is fully assembled at a Caterpillar facility following our quality guidelines
- Each generator set package is tested before leaving the Caterpillar facility
- Cat product support, including dealer service, parts and warranty covers the entire Cat power system

World Wide Product Support

- Cat dealers have over 1,800 dealer branch stores operating in 200 countries
- Your local Cat dealer provides extensive post-sale support, including maintenance and repair agreements

Standard Equipment

Air Inlet System

- Air cleaner; single element canister type with service indicator

Exhaust System

- Exhaust flange outlet
- Dry Exhaust Manifold
- Flanged faced outlet(s)

Generator

- IP23 Protection

- Matched to the performance and output characteristics of Cat engines

Power Termination

- Bus Bar

Control Panel

- EMCP 4 Genset Controller

Control System

- Alarm acknowledge
- kW-hr (total)
- Lamp test
- Over / under voltage (27 / 59)
- Digital indication for:
 - DC volts
 - PLC functionality
 - Speed adjust
- 4 Programmable relay outputs (Form A)♦
- Text alarm/event descriptions
- Warning / shutdown with common LED indication of shutdowns for:
 - Engine cycle crank
 - Emergency stop
 - Auto / start / stop control
- RPM
- Reverse Power (kW) (32)
- True RMS AC metering, 3-phase, +/-2% accuracy
- Accessory module data link
- Low coolant level
- kVAr-hr (total)
- kVAr (per phase, average & percent)
- 1 Programmable relay outputs (Form C)
- Over / under frequency (81 o/u)
- Low oil pressure
- Amps (per phase & average)
- Failure to start (overcrank)
- EMCP 4.2B
- Controls:
 - High coolant temperature
 - Communications
- Number of I/O varies on options selected
- 24 Volt DC operation

- Engine cool-down timer
- Low coolant temperature
- Generator phase sequence
- Frequency (Hz)
- Volts (L-L & L-N)
- Reverse reactive power (kVAr) (32RV)
- Oil pressure (psi, kPa or bar)
- Customizable screens
- Generator set packages include Caterpillar's Voltage Regulator (IVR). IVR includes reactive droop capability, 3-phase voltage sensing, KVAR / PF modes, RFI suppression, min max exciter. Limiter and exciter diode monitor
- Load histogram feature
- 1 Programmable digital outputs
- Customer data link (Modbus RTU)
- Overspeed
- Environmental sealed front face
- Coolant temperature
- kW (per phase, average & percent)
- Programmable protective relaying functions:
- kVA (per phase, average & percent)
- Power factor (per phase & average)
- Emergency stop pushbutton
- Generator mounted - rear facing
- Cat Connect
- Overcurrent (50 / 51)

Cooling System

- Package mounted radiator
- Radiator fan and fan drive
- Fan and belt guards
- Coolant drain line with valve
- Caterpillar Extended Life Coolant (Not included with packages without radiator, installed SF44CTS)
- Coolant level sensors (Not included with packages without radiator)
- SCAC

Fuel System

- Primary fuel filter with integral water separator
- Secondary fuel filters
- Flexible fuel lines - shipped loose
- Fuel priming pump
- Fuel cooler; not included with packages without radiator

Generators and Attachments

- Low voltage:
- Random wound
- 6 Leads
- NEMA standard hole pattern
- High Voltage
- Bus bar connections, right side extension box, bottom cable entry
- NEMA Class H insulation, Class H temperature rise at 40C ambient (125C prime / 150C standby)
- Permanent magnet
- NEMA standard hole pattern
- Medium voltage:
- Bus bar connections, top center mounted, top cable entry
- NEMA Class H insulation, Class H temperature rise at 40C ambient (125C prime / 150C standby)
- Winding temperature detectors
- Form wound
- 3 Phase brushless, salient pole
- Class H insulation, Class F temperature rise at 40C ambient (105C prime / 130C standby)
- Internal excitation

Governing System

- ADEM III

Literature

- English

Lube System

- Lubricating oil
- Gear type lube oil pump
- Oil drain lines and valve
- Fumes disposal
- Integral lube oil cooler
- Oil filter, filler and dipstick

Mounting System

- Rubber anti-vibration mounts (shipped loose)

Starting / Charging System

- Battery disconnect switch
- Battery and battery rack w/cables
- 45 Amp charging alternator
- 24 Volt electric starting motor

General

- Paint: Caterpillar yellow with high gloss black rails & radiator

- Flywheel and Flywheel housing-SAE No. 00
- Right hand servicing
- SAE Standard Rotation

Optional Equipment

Control Panels

- Remote monitoring software
- Digital I/O module
- EMCP 4.3, EMCP 4.4
- Load share module
- Generator temperature monitoring & protection

Exhaust System

- Exhaust mufflers
- Flanges
- Elbows
- Flange and exhaust expanders
- Mufflers
- Y Adapters
- Flexible fittings

Generator

- Oversize and premium generators
- Permanent magnet excitation (PMG)
- Internal excitation (IE)
- Anti-condensation heater

Air Inlet System

- Heavy duty air cleaner
- Single element filter

Power Termination

- Circuit breakers, UL listed
- Circuit breakers, IEC compliant

Cooling System

- Standard installed radiators
- Optional installed radiators
- Coolant
- Water level switch gauges

Fuel System

- Primary fuel filter

General

- UL 2200 package
- EEC Declaration of Conformity
- Seismic Certification per applicable building codes: IBC 2000, IBC 2003, IBC 2006, IBC 2009, CBC 2007
- CSA Certification
- Enclosures: sound attenuated, weather protective
- Automatic transfer switches (ATS)
- Integral & sub-base fuel tanks
- The following options are based on regional and product configuration:
- US aid emblem
- Integral & sub-base UL listed dual wall fuel tanks
- EU Certificate of Conformance (CE)

Generators and Attachments

- 1400, 1600, 2700 Frames
- Generator conversion from IE to PM
- Space heater
- RW, IE, 6 leads, Pitch 0.6667
- Low Voltage: 380,440 / 480 Volts, 3 Phase, 1800 rpm
- Differential current transformers (DCT) 15 kV class
- Generator air cleaner
- Differential current transformers (DCT) 8.7 kV class

Mounting

- IBC isolators
- Rubber anti-vibration mounts
- Spring-type vibration isolator

Starting/Charging

- Air starting motor with control and silencer
- Heavy-duty starting system
- Charging alternator
- Oversize batteries
- Battery chargers
- Jacket water heater

Instrumentation

- Wiring GP W/O cyl sensors
- Pyrometer and thermocouples

Lube System

- Lubrication oil in standard sump
- Oil drains
- Prelube pumps

- Oil level regulator

Crankcase Protection

- Explosion relief valve
- No crankcase explosion relief valve
- Crankcase ventilation system
- No explosion relief valves

Mounting System

- IBC vibration isolators
- Puck style low efficiency isolators
- Spring type vibration isolators

Power Connections

- 2000, 2500 Amp, 4 Pole, IEC Rated Circuit Breakers
- Cable entry - Right side
- Grounding groups
- Left side power connections
- 1600 ,2000, 2500, 3000, 4000 Amp, 3 Pole, UL-100% rated circuit breakers
- 2000, 2500,3200 Amp, 3 Pole, IEC Rated Circuit Breakers
- Rear power connections
- Neutral ground connections
- 6 Leads, 6 terminal option
- Right side power connections
- Center post buss bar

Special Tests / Reports

- Generator test report
- PGS test report @ 0.8 power factor
- Standard engine test charge
- CSA certification
- Torsional vibration analysis
- Generator fuel consumption test
- PGS test report @ 1.0 power factor
- IBC seismic certification
- Fuel consumption test

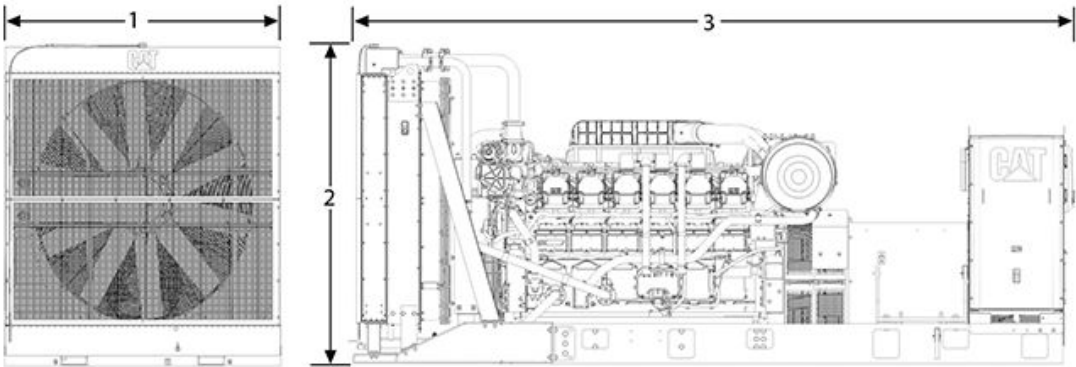
Starting / Charging System

- Air pressure regulator
- Batteries, racks and cables:
- Air starting motor
- 35 Amp battery charger
- 24 Volt battery set - Dry



- Starter covers
- 10 Amp battery charger
- Engine barring device
- Jacket water heaters
- Electric starting motors

Dimensional Art



Dimensions	Dimension 1	Dimension 2	Dimension 3
Genset dimensions	5637 mm (221.9 in)	2286 mm (90.0 in)	2367 mm (93.2 in)

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3512B

1230 ekW/ 1537 kVA/ 60 Hz/ 1800 rpm/ 480 V/ 0.8 Power Factor

Rating Type: CONTINUOUS

Fuel Strategy: LOW FUEL CONSUMPTION



Image shown may not reflect actual configuration

3512B

1230 ekW/ 1537 kVA
60 Hz/ 1800 rpm/ 480 V

Metric

English

Package Performance

Genset Power Rating with Fan @ 0.8 Power Factor	1230 ekW	
Genset Power Rating	1537 kVA	
Aftercooler (Separate Circuit)	30.0 ° C	86.0 ° F

Fuel Consumption

100% Load with Fan	325.1 L/hr	85.9 gal/hr
75% Load with Fan	243.4 L/hr	64.3 gal/hr
50% Load with Fan	171.6 L/hr	45.3 gal/hr
25% Load with Fan	108.0 L/hr	28.5 gal/hr

Cooling System¹

Engine Coolant Capacity	N/A	N/A
Radiator Water Capacity High Temp Circuit	N/A	N/A
Radiator Water Capacity Low Temp Circuit	N/A	N/A
Radiator Total Capacity	N/A	N/A

Inlet Air

Combustion Air Inlet Flow Rate	117.5 m³/min	4148.9 cfm
Max. Allowable Combustion Air Inlet Temp	49 ° C	120 ° F

Exhaust System

Exhaust Stack Gas Temperature	410.3 ° C	770.5 ° F
Exhaust Gas Flow Rate	282.2 m³/min	9964.4 cfm
Exhaust System Backpressure (Maximum Allowable)	6.7 kPa	27.0 in. water

3512B

1230 ekW/ 1537 kVA/ 60 Hz/ 1800 rpm/ 480 V/ 0.8 Power Factor

Rating Type: CONTINUOUS

Fuel Strategy: LOW FUEL CONSUMPTION

Heat Rejection

Heat Rejection to Jacket Water	519 kW	29515 Btu/min
Heat Rejection to Exhaust (Total)	1172 kW	66649 Btu/min
Heat Rejection to Aftercooler	355 kW	20188 Btu/min
Heat Rejection to Atmosphere from Engine	113 kW	6426 Btu/min
Heat Rejection to Atmosphere from Generator	58 kW	3298 Btu/min

Alternator²

Motor Starting Capability @ 30% Voltage Dip	4350 skVA
Current	1849 amps
Frame Size	1447
Excitation	IE
Temperature Rise	105 ° C

Emissions (Nominal)³

NOx	2858.7 mg/Nm ³	6.1 g/hp-hr
CO	675.6 mg/Nm ³	1.4 g/hp-hr
HC	186.2 mg/Nm ³	0.4 g/hp-hr
PM	50.3 mg/Nm ³	0.1 g/hp-hr

DEFINITIONS AND CONDITIONS

1. For ambient and altitude capabilities consult your Cat dealer. Air flow restriction (system) is added to existing restriction from factory.
2. UL 2200 Listed packages may have oversized generators with a different temperature rise and motor starting characteristics. Generator temperature rise is based on a 40° C ambient per NEMA MG1-32.
3. Emissions data measurement procedures are consistent with those described in EPA CFR 40 Part 89, Subpart D & E and ISO8178-1 for measuring HC, CO, PM, NOx. Data shown is based on steady state operating conditions of 77° F, 28.42 in HG and number 2 diesel fuel with 35° API and LHV of 18,390 btu/lb. The nominal emissions data shown is subject to instrumentation, measurement, facility and engine to engine variations. Emissions data is based on 100% load and thus cannot be used to compare to EPA regulations which use values based on a weighted cycle.

Applicable Codes and Standards:

AS1359, CSA C22.2 No100-04, UL142, UL489, UL869, UL2200, NFPA37, NFPA70, NFPA99, NFPA110, IBC, IEC60034-1, ISO3046, ISO8528, NEMA MG1-22, NEMA MG1-33, 2006/95/EC, 2006/42/EC, 2004/108/EC.

Note: Codes may not be available in all model configurations. Please consult your local Cat Dealer representative for availability.

3512B

1230 ekW/ 1537 kVA/ 60 Hz/ 1800 rpm/ 480 V/ 0.8 Power Factor

Rating Type: CONTINUOUS

Fuel Strategy: LOW FUEL CONSUMPTION

CONTINUOUS: Output available with non-varying load for an unlimited time. Average power output is 70-100% of the continuous power rating. Typical peak demand is 100% of continuous rated ekW for 100% of operating hours.

Ratings are based on SAE J1349 standard conditions. These ratings also apply at ISO3046 standard conditions

Fuel Rates are based on fuel oil of 35° API [16° C (60° F)] gravity having an LHV of 42 780 kJ/kg (18,390 Btu/lb) when used at 15° C (59° F) and weighing 850 g/liter (7.094 lbs/U.S. gal.). Additional ratings may be available for specific customer requirements, contact your Cat representative for details. For information regarding Low Sulfur fuel and Biodiesel capability, please consult your Cat dealer.

Performance No.: DM8188-01

Feature Code: 512DRC4

Generator Arrangement: 2523806

Date: 03/15/2021

Source Country: U.S.

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